Examiner only

10. Halogenoalkanes are hydrolysed when heated with aqueous sodium hydroxide.

(a) (i) Write the equation to show the hydrolysis of 3-chloro-2-methylpentane with aqueous sodium hydroxide. You should show clearly the structure of the organic reagent and product. [1]

(ii) State the name of the mechanism for the reaction taking place in part (i). [1]

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(iii) Describe a chemical test to show that this hydrolysis reaction has occurred. Include the test and the result expected. [2]

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(b) The rate of reaction between a halogenoalkane and aqueous sodium hydroxide was measured when different concentrations of reagents were used. The results are shown in the table.

Experiment	Initial concentration OH^-(aq) / mol dm^{-3}	Initial concentration halogenoalkane / mol dm^{-3}	Initial rate of reaction / mol dm^{-3} s^{-1}
1	0.10	0.10	1.20×10^{-5}
2	0.10	0.20	2.40×10^{-5}
3	0.10	0.30	3.60×10^{-5}
4	0.20	0.10	1.20×10^{-5}
5	0.30	0.10	1.20×10^{-5}

(i) Use these data to deduce how the concentration of each reagent affects the rate of reaction. Explain how you reached your conclusions. [2]

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(ii) What would be the effect on the initial rate of reaction if the halogen in the halogenoalkane were changed from chlorine to bromine? You should assume that this is the only change made. Explain your answer. [3]

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